

Basic Linux Commands – Complete Guide

◇ Introduction

Linux commands terminal mein kaam karne ki foundation hain. In commands ki madad se aap files, directories, processes, storage, network aur system ko manage kar sakte hain. Agar aap Termux, Ubuntu, Kali Linux ya kisi bhi Linux distribution ka use karte hain, to ye commands sabse basic aur important hain.

FILE AND DIRECTORY COMMANDS

1. pwd – Print Working Directory

Ye command current working directory ka complete path dikhati hai.

Command

```
pwd
```

Example Output

```
/data/data/com.termux/files/home
```

2. `ls` – List Directory Contents

Ye command current directory ki files aur folders display karti hai.

Common Options

Sirf files aur folders dekhne ke liye:

```
ls
```

Detailed information dekhne ke liye:

```
ls -l
```

Hidden files dekhne ke liye:

```
ls -a
```

Detailed aur hidden files ek saath dekhne ke liye:

```
ls -la
```

Human-readable file sizes (KB, MB, GB) ke saath:

```
ls -lh
```

Time ke hisaab se sorted list:

```
ls -lt
```

Example

```
ls -la
```

Permission Symbols

`d` ka matlab Directory hota hai.

`-` ka matlab Normal File hota hai.

l ka matlab Symbolic Link hota hai.

r = Read Permission

w = Write Permission

x = Execute Permission

Permissions teen groups mein hoti hain: **Owner, Group aur Others.**

3. cd – Change Directory

Ye command current directory ko change karti hai.

Specific directory mein jane ke liye:

```
cd <directory>
```

Ek level upar jane ke liye:

```
cd ..
```

Do levels upar jane ke liye:

```
cd ../..
```

Root directory mein jane ke liye:

```
cd /
```

Home directory mein jane ke liye:

```
cd ~
```

Previous directory mein wapas jane ke liye:

```
cd -
```

Example

```
cd storage/downloads  
cd ..  
cd ~
```

4. mkdir – Make Directory

Naya folder create karne ke liye:

```
mkdir myfolder
```

Nested directories create karne ke liye:

```
mkdir -p project/src/assets
```

5. rmdir – Remove Directory

Empty directory delete karne ke liye:

```
rmdir myfolder
```

6. rm – Remove Files and Directories

Single file delete karne ke liye:

```
rm file.txt
```

Directory delete karne ke liye:

```
rm -r myfolder
```

Force delete karne ke liye:

```
rm -rf project/
```

Delete se pehle confirmation lene ke liye:

```
rm -i file.txt
```

Note: rm -rf permanently data delete karta hai.

7. cp – Copy Files and Directories

File copy karne ke liye:

```
cp file.txt /sdcard/
```

Directory copy karne ke liye:

```
cp -r documents/ backups/
```

Confirmation ke saath overwrite karne ke liye:

```
cp -i file.txt backup.txt
```

8. mv – Move ya Rename Files

File move karne ke liye:

```
mv file.txt /sdcard/
```

File rename karne ke liye:

```
mv oldname.txt newname.txt
```

9. touch – Empty File Create Karna

```
touch newfile.txt
```

Ye command nayi empty file create karti hai ya existing file ka timestamp update karti hai.

10. cat – File Content Dekhna

File ka content dekhne ke liye:

```
cat file.txt
```

Nayi file create ya edit karne ke liye:

```
cat > newfile.txt
```

Multiple files combine karne ke liye:

```
cat file1 file2 > combined.txt
```

11. more aur less

Large files ko page-by-page dekhne ke liye:

```
more largefile.txt
```

Advanced viewer:

```
less largefile.txt
```

12. head aur tail

File ki pehli 10 lines dekhne ke liye:

```
head -n 10 file.txt
```

Aakhri 20 lines dekhne ke liye:

```
tail -n 20 file.txt
```

Real-time log monitoring ke liye:

```
tail -f log.txt
```

13. find – Files Search Karna

Name ke basis par search:

```
find . -name "*.txt"
```

Sirf files search karne ke liye:

```
find /sdcard -type f
```

Sirf directories search karne ke liye:

```
find /sdcard -type d
```

Size ke basis par search:

```
find /sdcard -size +50M
```

14. grep – Text Search

File ke andar text search karne ke liye:

```
grep "error" log.txt
```

Case-insensitive search:

```
grep -i "error" log.txt
```

Recursive search:

```
grep -r "password" /sdcard/
```

Line number ke saath search:

```
grep -n "error" log.txt
```

FILE PERMISSION COMMANDS

15. chmod

Ye command file ya directory ki permissions change karti hai.

Permission values:

4 = Read

2 = Write

1 = Execute

7 = Read + Write + Execute

6 = Read + Write

5 = Read + Execute

3 = Write + Execute

Examples

```
chmod 755 script.sh
```

```
chmod 644 file.txt
```

```
chmod +x script.sh
```

```
chmod -x script.sh
```

16. chown

File ka owner change karne ke liye:

```
chown user:group file.txt
```

Is command ke liye generally root privileges ki zaroorat hoti hai.

DISK AND STORAGE COMMANDS

17. df

Disk space dekhne ke liye:

df -h

Specific partition:

df -h /

18. du

Directory ka total size:

du -sh /sdcard/

Har subdirectory ka size:

du -h storage/

SYSTEM INFORMATION COMMANDS

19. whoami

Current user dekhne ke liye:

whoami

20. uname

Complete system information:

uname -a

Kernel version:

`uname -r`

Architecture:

`uname -m`

21. `date`

Current date aur time:

`date`

22. `uptime`

System kitni der se chal raha hai:

`uptime`

23. `ps`

Running processes dekhne ke liye:

`ps`

Sabhi processes:

`ps aux`

Specific process search:

`ps aux | grep python`

24. top

Real-time process monitor:
top

Exit karne ke liye q press karein.

25. kill

Process terminate karne ke liye:
kill 12345

Force terminate karne ke liye:
kill -9 12345

Name ke basis par process terminate karne ke liye:
pkill process_name

NETWORK COMMANDS

26. ping

Internet connectivity check karne ke liye:
ping google.com

Sirf 4 packets bhejne ke liye:
ping -c 4 8.8.8.8

Stop karne ke liye Ctrl + C press karein.

27. Network Information

Network interfaces:

```
ifconfig
```

IP address dekhne ke liye:

```
ip addr
```

28. wget

Internet se file download karne ke liye:

```
wget <url>
```

Custom filename ke saath:

```
wget -O filename.zip <url>
```

29. curl

Web content retrieve karne ke liye:

```
curl <url>
```

File download karne ke liye:

```
curl -o file.zip <url>
```

PACKAGE MANAGEMENT

30. pkg (Termux)

Package list update:

`pkg update`

Packages upgrade:

`pkg upgrade`

Package install:

`pkg install <package>`

Package remove:

`pkg uninstall <package>`

Package search:

`pkg search <package>`

Installed packages:

`pkg list-installed`

31. apt (Debian/Ubuntu/Kali)

Package list update:

`apt update`

Packages upgrade:

`apt upgrade`

Package install:

`apt install <package>`

Package remove:

`apt remove <package>`

Package search:

`apt search <package>`

ARCHIVE AND COMPRESSION

32. tar

Archive create karne ke liye:

`tar -cvf archive.tar files`

Archive extract karne ke liye:

`tar -xvf archive.tar`

Compressed archive extract karne ke liye:

`tar -xzvf archive.tar.gz`

33. zip aur unzip

ZIP archive create karne ke liye:

`zip archive.zip file1.txt file2.txt`

Extract karne ke liye:

`unzip archive.zip`

Specific directory mein extract karne ke liye:

`unzip archive.zip -d /sdcard/`

REDIRECT AND PIPE

Output overwrite karne ke liye:

```
echo "Hello" > file.txt
```

Output append karne ke liye:

```
echo "World" >> file.txt
```

Pipe operator ka use:

```
ls -la | grep "txt"
```

```
cat file.txt | grep "error"
```

TEXT EDITORS

34. Nano Editor

```
nano file.txt
```

Useful shortcuts:

- Ctrl + O : Save
- Ctrl + X : Exit
- Ctrl + K : Cut Line
- Ctrl + U : Paste

35. Vim Editor

```
vim file.txt
```

Useful shortcuts:

- i : Insert Mode
- Esc : Command Mode

- :w : Save
- :q : Quit
- :wq : Save and Quit

◇ **ADDITIONAL USEFUL COMMANDS**

Command history dekhne ke liye:

history

Terminal screen clear karne ke liye:

clear

⚠ **IMPORTANT NOTE**

Ye notes sirf educational aur learning purposes ke liye hain.

Linux commands ka istemal sirf un systems aur devices par

karein jinke liye aapke paas authorization ho. Bina

permission ke access ya misuse karna applicable cyber laws ke under punishable offense ho sakta hai.